Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam D · Form 2: Multi-step reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. A target is 4 m horizontally away and 1.225 m below a horizontal launcher. Ignore drag. Find time to reach the target. Use g = 9.8 m/s².

(A) 0.5 s

(B) -0.5 s

(C) 1 s

(D) 0.25 s

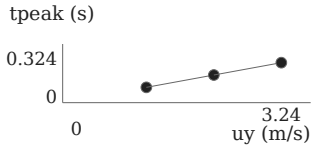
2. In gravity g=10 m/s² vary the upward component of launch velocity and time the ascent. Double gravitational acceleration. What is the new ratio of vertical change to the same horizontal change?

(A) 0.14

(B) 0.05

(C) 0.08

(D) -0.01



3. A robot moves 6 m east, 14 m north, then 4 m west. What is the angle of its displacement north of east?

(A) 81.87 °

(B) -81.87 °

(C) 163.74 °

(D) 40.935 °

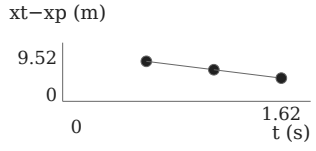
4. A projectile moves horizontally at +6 m/s while a target travels at +2 m/s. Track their horizontal separation during flight. t (s) -> xt-xp (m): (0.62, 9.52); (1.12, 7.52); (1.62, 5.52). Which comparison distinguishes vertical zero bias from slope error?

(A) Remove units before comparing.

(B) Force every fit through the origin.

(C) Compare observed minus predicted residuals at separated settings.

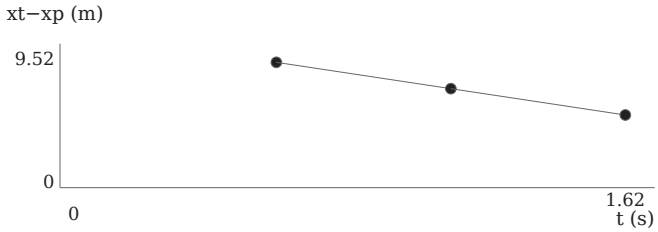
(D) Repeat at only one setting.



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A projectile moves horizontally at +6 m/s while a target travels at +2 m/s. Track their horizontal separation during flight. t (s) -> xt-xp (m): (0.62, 9.52); (1.12, 7.52); (1.62, 5.52).

a. For the moving target setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Hold the target at its initial position. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. A ball launches horizontally from a 1.25 m table. Use g=10 m/s² and neglect drag. vx (m/s) -> R (m): (1.24, 0.62); (2.24, 1.12); (3.24, 1.62).

a. Build the model for horizontal-launch target from physics and predict both slope and intercept. **(2 pt)**

b. Quadruple the table height. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. In gravity g=10 m/s² vary the upward component of launch velocity and time the ascent. uy (m/s) -> tpeak (s): (1.24, 0.124); (2.24, 0.224); (3.24, 0.324). Build the model for apex timing from physics and predict both slope and intercept. **(3 pt)**